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Giovanni Bordiga

Curriculum Vitae

Research interests

Mechanical intelligence · Inverse design · Differentiable physics · Optimization · Mechanical metamaterials · Material instabilities · Wave propagation · Homogenization theory

Education

Nov. 2016 – Apr. 2020 **PhD in Solid and Structural Mechanics**,
Doctoral School of Civil, Environmental, and Mechanical Engineering,
University of Trento (Italy),
Curriculum: Modeling and Simulation

- PhD Thesis: *"Homogenization of periodic lattice materials for wave propagation, localization, and bifurcation"*,
  Advisors: Andrea Piccolroaz and Davide Bigoni,
  Evaluation Commitee: Basile Audoly and Pedro M. Reis,
  Final grade: *Cum Laude*.

Sep. 2014 – Oct. 2016 **MSc in Civil Engineering**,
Department of Civil, Environmental, and Mechanical Engineering,
University of Trento (Italy),
Curriculum: Structures

- MSc Thesis: *"Micromechanical modeling of masonry"* (in Italian),
  Advisor: Davide Bigoni,
  Final grade: *110/110 Cum Laude*.

Sep. 2011 – Sep. 2014 **BSc in Civil Engineering**,
Department of Civil, Environmental, and Mechanical Engineering,
University of Trento (Italy)

- BSc Thesis: *"Non-uniform torsion of thin-walled open cross-section beams"* (in Italian),
  Advisor: Massimiliano Gei,
  Final grade: *110/110 Cum Laude*.

Research experience

Sep. 2021 – present **Postdoctoral Fellow**,
Harvard School of Engineering and Applied Sciences,
Harvard University (US),
Advisors: Katia Bertoldi

- Development of a differentiable simulation framework for automating the design of flexible mechanical metamaterials with target dynamic responses.
- Inverse design of flexible mechanical metamaterials for shape morphing, nonlinear wave guiding, energy focusing, shock mitigation, and cloaking of nonlinear waves.

May 2020 – Aug. 2021 **Postdoctoral Researcher**,
Department of Civil, Environmental, and Mechanical Engineering,
University of Trento (Italy),
Advisors: Andrea Piccolroaz and Davide Bigoni

- Homogenization of elastic lattices prestressed by circulatory (follower) forces demonstrating the existence of hypoelastic continuum materials displaying flutter instability at the macroscale.
- Analytical and numerical investigation of tensile buckling in periodic lattice materials endowed with sliding constraints.

Nov. 2016 – Apr. 2020 **PhD Student in Solid and Structural Mechanics**,
Department of Civil, Environmental, and Mechanical Engineering,
University of Trento (Italy),
Advisors: Andrea Piccolroaz and Davide Bigoni

- Analytical and numerical investigation of the free and forced dynamic response of Rayleigh-beam lattices leading to sharp frequency-dependent wave localization.
- Modeling and tuning of the prestress state in a lattice domain to realize dynamic interfaces designed for total reflection, negative refraction, and wave channeling.
- Development of static and dynamic homogenization methods for beam lattices prestressed by conservative and non-conservative positional forces.
- Analytical and numerical analysis of macroscopic and microscopic bifurcations in prestressed anisotropic lattices and their connection to shear bands formation and short-wavelength instabilities.

Entrepreneurial experience

Apr., 2024 – present **Founder & Chief Technology Officer (CTO)**,
SoftPulse, Inc., Cambridge, MA (US),
Mission: Develop a passive implant for lasting blood pressure control. A drug-free solution for millions with hypertension.
Team: Adel Djellouli, Giovanni Bordiga, and Katia Bertoldi.
Funding: $160K from the Harvard Grid Accelerator.

Publications

Preprints

[2] A. A. Watkins, **G. Bordiga**, V. Tournat, K. Bertoldi, *Wave-based reading of mechanical memory in multistable mass-in-mass metamaterials*. 2026. DOI: 10.48550/arXiv.2609.07889. Pre-published.

[1] N. Singh, A. A. Watkins, **G. Bordiga**, V. Tournat, K. Bertoldi, Z. Rocklin, *Conformal Elastodynamics in 2D Dilational Metamaterials*. 2026. DOI: 10.48550/arXiv.2604.16637. (Accepted in *Nat. Phys.*) Pre-published.

Peer-reviewed journals

[11] A. A. Watkins, **G. Bordiga**, M. Mu, V. Tournat, K. Bertoldi, "Arbitrary mechanical memory encoding via nonlinear waves in bistable metamaterials". *Phys. Rev. Lett.* 137.9 (2026), p. 097201. DOI: 10.1103/pkk5-dykb.

[10] **G. Bordiga**, J.-G. Argaud, A. A. Watkins, V. Tournat, K. Bertoldi, "Nonlinear mechanical metamaterial cloaks". *Adv. Funct. Mater.* (2025), e22895. DOI: 10.1002/adfm.202522895.

[9] C. Perez-Garcia, R. Zaera, J. Aranda-Ruiz, **G. Bordiga**, G. Risso, M. L. Lopez-Donaire, K. Bertoldi, D. Garcia-Gonzalez, "Reprogrammable mechanical metamaterials via passive and active magnetic interactions". *Adv. Mater.* (2025), p. 2412353. DOI: 10.1002/adma.202412353.

[8] A. S. Meeussen, **G. Bordiga**, A. X. Chang, B. Spoettling, K. P. Becker, L. Mahadevan, K. Bertoldi, "Textile hinges enable extreme properties of kirigami metamaterials". *Adv. Funct. Mater.* (2024), p. 2415986. DOI: 10.1002/adfm.202415986.

[7] **G. Bordiga**, E. Medina, S. Jafarzadeh, C. Bösch, R. P. Adams, V. Tournat, K. Bertoldi, "Automated discovery of reprogrammable nonlinear dynamic metamaterials". *Nat. Mater.* 23.11 (2024), pp. 1486–1494. DOI: 10.1038/s41563-024-02008-6.

[6] A. Kotikian, A. A. Watkins, **G. Bordiga**, A. Spielberg, Z. S. Davidson, K. Bertoldi, J. A. Lewis, "Liquid crystal elastomer lattices with thermally programmable deformation via multi-material 3D printing". *Adv. Mater.* 36.34 (2024), p. 2310743. DOI: 10.1002/adma.202310743.

[5] **G. Bordiga**, D. Bigoni, A. Piccolroaz, "Tensile material instabilities in elastic beam lattices lead to a bounded stability domain". *Philos. Trans. R. Soc. Math. Phys. Eng. Sci.* 380.2231 (2022), p. 20210388. DOI: 10.1098/rsta.2021.0388. (Cover article).

[4] **G. Bordiga**, A. Piccolroaz, D. Bigoni, "A way to hypo-elastic artificial materials without a strain potential and displaying flutter instability". *J. Mech. Phys. Solids* 158 (2022), p. 104665. DOI: 10.1016/j.jmps.2021.104665.

[3] **G. Bordiga**, L. Cabras, A. Piccolroaz, D. Bigoni, "Dynamics of prestressed elastic lattices: Homogenization, instabilities, and strain localization". *J. Mech. Phys. Solids* 146 (2021), p. 104198. DOI: 10.1016/j.jmps.2020.104198.

[2] **G. Bordiga**, L. Cabras, D. Bigoni, A. Piccolroaz, "Free and forced wave propagation in a Rayleigh-beam grid: Flat bands, Dirac cones, and vibration localization vs isotropization". *Int. J. Solids Struct.* 161 (2019), pp. 64–81. DOI: 10.1016/j.ijsolstr.2018.11.007.

[1] **G. Bordiga**, L. Cabras, A. Piccolroaz, D. Bigoni, "Prestress tuning of negative refraction and wave channeling from flexural sources". *Appl. Phys. Lett.* 114.4 (2019), p. 041901. DOI: 10.1063/1.5084258. (Editor's Pick).

In conference proceedings

[3] **G. Bordiga**. "In-plane Floquet-Bloch waves in elastic grids and prestress tuning of structured interfaces". *RAMSS2019 - Recent Advances in Mechanics of Solids and Structures*. Trento, 2019. URL: http://rgdoi.net/10.13140/RG.2.2.15758.82244.

[2] **G. Bordiga**. "Free and forced wave propagation in a Rayleigh-beam grid: flat bands, Dirac cones, and vibration localization vs isotropization". *ESMC2018 - 10th European Solid Mechanics Conference*. Bologna, 2018. URL: http://rgdoi.net/10.13140/RG.2.2.24679.09126.

[1] **G. Bordiga**. "Micromechanical modelling of masonry". *CERMODEL2017*. Trento, 2017.

Software

[3] **G. Bordiga**. *DifFlexMM: Differentiable Flexible Mechanical Metamaterials*. Bertoldi Group, 2024. URL: https://github.com/bertoldi-collab/DifFlexMM.

[2] **G. Bordiga**. *Simulation and Design of Shape-Morphing LCE Lattices*. Bertoldi Group, 2024. URL: https://github.com/bertoldi-collab/morphing-lattices.

[1] **G. Bordiga**. *A humble image tracking code*. Bertoldi Group, 2023. URL: https://github.com/bertoldi-collab/tracking-markers.

Conference presentations

Mar. 16–20, 2026 — **APS2026 Global Physics Summit**, Denver, CO (US),
Tuning nonlinear mechanical metamaterials for fully-analog sensing and robotic control,
G. Bordiga, C. Bösch, C. McCann, M. Yuen, J. Wilt, S. Jafarzadeh, K. Bertoldi.

Mar. 16–21, 2025 — **APS2025 Global Physics Summit**, Anaheim, CA (US),
Nonlinear mechanical metamaterial cloaks,
G. Bordiga, J. Argaud, A. A. Watkins, V. Tournat, K. Bertoldi.

Sep. 9–11, 2024 — **SMASIS2024**, Atlanta, GA (US),
Automated design of flexible mechanical metamaterials with reprogrammable wave functionalities,
G. Bordiga, E. Medina, S. Jafarzadeh, C. Bösch, V. Tournat, K. Bertoldi.

Sep. 9–11, 2024 — **SMASIS2024**, Atlanta, GA (US),
Mechanical metamaterial 'brain' for fully analog control of a mobile robot,
C. Bösch, **G. Bordiga**, C. McCann, S. Jafarzadeh, J. Wilt, M. Yuen, Y. Jin, A. Fichtner, K. Bertoldi.

Jul. 3–4, 2024 **DynaMetaFlex**, *Workshop on "Nonlinear dynamics of flexible mechanical metamaterials"*, Laboratoire d'Acoustique de l'Université du Mans, Institut d'Acoustique – Graduate School, CNRS, Le Mans Université, Le Mans, (France),
Teaching reprogrammable dynamics to material structures,
G. Bordiga, E. Medina, S. Jafarzadeh, C. Bösch, V. Tournat, K. Bertoldi.

Oct. 8–11, 2023 **SES2023**, Minneapolis, MN (US),
Inverse-design of nonlinear mechanical metamaterial cloaks,
G. Bordiga, J. Argaud, V. Tournat, K. Bertoldi.

Mar. 5–10, 2023 **APS2023**, Las Vegas, NV (US),
Manipulating energy flows with non-periodic mechanical metamaterials,
G. Bordiga, E. Medina, V. Tournat, K. Bertoldi.

Oct. 16–19, 2022 **SES2022**, College Station, TX (US),
Non-periodic design discovery for optimal dynamic responses in flexible mechanical metamaterials,
G. Bordiga, E. Medina, V. Tournat, K. Bertoldi.

Jul. 4–8, 2022 **ESMC2022**, *11th European Solid Mechanics Conference*, Galway (Ireland),
Non-hyper-elastic materials from follower prestress states reveal flutter instability in elastic lattices (Invited talk),
G. Bordiga, A. Piccolroaz, D. Bigoni.

Mar. 14–18, 2022 **APS2022**, Chicago, IL (US),
Architected metamaterials for routing nonlinear mechanical pulses,
G. Bordiga, E. Medina, V. Tournat, K. Bertoldi.

Invited talks & Seminars

Jun. 22–25, 2026 **NSF ID4 Workshop**, *Accelerating Quantum Materials Discovery Workshop*, Colorado School of Mines, Golden, CO (US),
Nonlinear mechanical metamaterial cloaks,
G. Bordiga, J. Argaud, A. A. Watkins, V. Tournat, K. Bertoldi.

Mar. 31, 2025 **Seminar**, *Center for Fluid Mechanics Seminar*, Brown University, Providence, RI (US),
Automating the discovery of nonlinear architected materials,
G. Bordiga.

Jul. 25, 2024 **Seminar**, *Seismology and Wave Physics Lab*, ETH, Zürich (Switzerland),
Teaching reprogrammable dynamics to material structures,
G. Bordiga.

Jun. 13, 2024 **Seminar**, *Laboratory for Intelligent Probabilistic Systems*, Princeton University, Princeton, NJ (US),
Teaching reprogrammable dynamics to material structures,
G. Bordiga.

Jul. 4–8, 2022 **ESMC2022**, *11th European Solid Mechanics Conference*, Galway (Ireland),
Non-hyper-elastic materials from follower prestress states reveal flutter instability in elastic lattices,
G. Bordiga, A. Piccolroaz, D. Bigoni.

Media coverage

About my research

Sep. 15, 2025 **Colorado School of Mines**, *The Institute for Data Driven Dynamical Design (ID4)*,
ID4 Research Highlights on Dynamic Metamaterials

May 7, 2025 **Azo Materials**,
Novel Mechanical Metamaterials with Magnetically Controlled Properties

May 6, 2025 **Tech Xplore**,
Structurally reprogrammable magnetic metamaterials hold promise for biomedicine, soft robotics

Oct. 17, 2024 **Nature Materials News and Views**, *Gary P. T. Choi*,
Designing flexible mechanical metamaterials with complex functionalities

Oct. 31, 2024 **Communication CNRS Ingénierie**,
La conception de métamatériaux guidée par des outils de l'IA

About SoftPulse, Inc.

Jul. 8, 2025 **The Harvard Gazette**, *Kirsten Mabry, Harvard Office of Technology Development*,
3 tech solutions to societal needs will get help moving to market

Mentoring

At Bertoldi Lab, Harvard University

Sep., 2025 – present **Yanzi Wang**, *Undergraduate student*, **Tsinghua University (China)**,
Project: *Mechanical metamaterials for underactuated dynamic path generation*.

Sep., 2024 – present **Louis-Justin Tallot**, *PhD student*, **Harvard University (US)**,
Project: *Differentiable origami simulation; Inverse-design of magneto-mechanical metamaterials*.

Sep., 2024 – Dec., 2025 **Stijn de Bruin**, *Master student*, **TU Delft (Netherlands)**,
Project: *Encoding dynamic bifurcations in mechanical metamaterials*.

Sep., 2024 – Jan., 2025 **Antoine Fondeur**, *Master student*, **MINES Paris – PSL (France)**,
Project: *Inverse design of magneto-mechanical metamaterials for reprogrammable static and dynamic responses*.

Sep., 2024 – Jan., 2025 **Jian Zhimo**, *PhD student*, **Tsinghua University (China)**,
Project: *Mechanical metamaterials for sensing and control*.

Sep., 2023 – Sep., 2024 **Tom Vreugdenhil**, *Master student*, **TU Delft (Netherlands)**,
Thesis: *Dynamic metamaterials for reprogrammable underactuated robotics*.

Sep., 2023 – Dec., 2024 **Carlos Peréz García**, *PhD student*, **University Carlos III of Madrid (Spain)**,
Project: *Magneto-mechanical metamaterials for reprogrammable stress-strain response and energy absorption*.

Feb.–Sep., 2023 **Jean-Gabriel Argaud**, *Master student*, **MINES Paris – PSL (France)**,
Project: *Cloaking in nonlinear mechanical metamaterials*.

Jul.–Dec., 2022 **Cyrill Bösch**, *PhD student*, **ETH (Switzerland)**,
Project: *Mechanical metamaterial 'brain' enables fully analog robotic control*.

Sep., 2022–present **Audrey Watkins**, *PhD student*, **Harvard University (US)**,
Project: *Nonlinear waves in multistable metamaterials for mechanical computing*.

Feb.–Sep., 2022 **Ben Spoettling**, *Master student*, **ETH (Switzerland)**,
Thesis: *Automated design and discovery of shape-morphing metamaterials*.

Teaching experience

2018–2020 **Teaching assistant**,
Department of Civil, Environmental, and Mechanical Engineering,
University of Trento (Italy),
Course: *Structural Dynamics*

- 40+ students.
- Material: Created interactive Mathematica notebooks to teach the dynamics of single and multi-degree-of-freedom systems, modal analysis, free and forced vibrations of elastic beams.
- Graded: oral exams.

2019–2020 **Lecturer and tutor**,
Department of Civil, Environmental, and Mechanical Engineering,
University of Trento (Italy),
Course: *Solids and Structural Mechanics*

- 80+ students, 20 hours of lectures, and 5 hours of tutoring activities.
- Taught: De Saint Venant theory and exercises, Principle of Virtual Work for statically indeterminate structures, structural symmetries, frame structures.

- Graded: written and oral exams.

2018–2019 **Lecturer and tutor**,
Department of Civil, Environmental, and Mechanical Engineering,
University of Trento (Italy),
Course: *Solids and Structural Mechanics*

- 80+ students, 20 hours of lectures, and 20 hours of tutoring activities.
- Taught: De Saint Venant theory and exercises, Principle of Virtual Work for statically indeterminate structures, Mohr's circles, elastic beam theory, frame structures.
- Graded: written and oral exams.

2017–2018 **Lecturer and tutor**,
Department of Civil, Environmental, and Mechanical Engineering,
University of Trento (Italy),
Course: *Solids and Structural Mechanics*

- 80+ students, 30 hours of lectures, and 30 hours of tutoring activities.
- Taught: De Saint Venant theory and exercises, Mohr's circles, failure and yield criteria (Von Mises, Tresca, Mohr-Coulomb, Rankine, Drucker-Prager), Principle of Virtual Work for statically indeterminate structures, elastic beam theory.
- Graded: written and oral exams.

Peer reviews

- Science Advances
- Journal of the Mechanics and Physics of Solids
- Advanced Engineering Materials
- International Journal of Solids and Structures
- Nonlinear Dynamics
- European Journal of Mechanics - A/Solids
- Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences
- Computer Methods in Applied Mechanics and Engineering
- Journal of Elasticity
- Mechanics of Materials
- Journal of Sound and Vibration
- Meccanica
- Journal of Mechanics of Materials and Structures
- ACS Photonics

Events & Conferences service

Mar. 15–20, 2026 **APS2026 Global Physics Summit**, Denver, CO (US),
Session: Functionality Through Nonlinearity
Served as: **Session chair**.

Mar. 15, 2026 **Squishy Science Sunday at APS2026 Global Physics Summit**, Denver, CO (US),
Activity: Marbelous Marbling
Served as: **Volunteer**.

Sep. 9–11, 2024 **SMASIS2024**, Atlanta, GA (US),
Special Symposium: Embodying Physical Computing and Mechano-Intelligence
Served as: **Session chair**.

Jul. 22, 2024 **MRSEC Workshop**, Harvard SEAS, Cambridge, MA (US),
Git + GitHub Workshop: Version control for scientists
Served as: **Organizer and presenter**.

Jul. 29, 2022 **MRSEC Workshop**, Harvard SEAS, Cambridge, MA (US),
Git + GitHub Workshop: Version control for scientists
Served as: **Organizer**.

Sep. 27, 2019 **European Researchers' night**, Trento (Italy)
Served as: **Co-organizer and presenter** for the Solid and Structural Mechanics exhibition stand.

Jun. 6–7, 2019 **RAMSS2019 — Recent Advances in Mechanics of Solids and Structures**, Trento (Italy)
Served as: **Conference assistant**.

Sep. 28, 2018 **European Researchers' night**, Trento (Italy)
Served as: **Co-organizer and presenter** for the Solid and Structural Mechanics exhibition stand.

Jul. 2–6, 2018 **ESMC2018 — 10th European Solid Mechanics Conference**, Bologna (Italy)
Served as: **Co-organizer** for conference program.

Sep. 29, 2017 **European Researchers' night**, Trento (Italy)
Served as: **Co-organizer and presenter** for the Solid and Structural Mechanics exhibition stand.

Awards & Honors

2024 $160K from the Harvard Grid Accelerator to de-risk and develop SoftPulse hypertension implant towards animal trials.

2022 Best PhD Thesis Award for Cycle XXXII, University of Trento (Italy)

2017 Graduation Award for Merit, University of Trento (Italy)

2016 University Scholarship for Merit, Italian government, INPS

2015 University Scholarship for Merit, Italian government, INPS

2015 Graduation Award for Merit, University of Trento (Italy)

2014 University Scholarship for Merit, Italian government, INPS

2013 University Scholarship for Merit, Italian government, INPS

2012 University Scholarship for Merit, Italian government, INPS

Community activities

Jun.–Aug., 2026 **Beach volleyball tournament**, *Harvard Rhino League*, Cambridge, MA (US)
Role: Team captain for the Bertoldi Lab (4th place).

Jun.–Aug., 2025 **Beach volleyball tournament**, *Harvard Rhino League*, Cambridge, MA (US)
Role: Team captain for the Bertoldi Lab (2nd place).

Jun.–Aug., 2024 **Beach volleyball tournament**, *Harvard Rhino League*, Cambridge, MA (US)
Role: Team captain for the Bertoldi Lab (Championship winner).

Jun.–Aug., 2023 **Beach volleyball tournament**, *Harvard Rhino League*, Cambridge, MA (US)
Role: Team captain for the Bertoldi Lab (Championship winner).

Skills

Programming

Scientific computing	Wolfram Language	*Highly specialized*	Python	*Highly specialized*
	MATLAB	*Highly specialized*	R	*Intermediate*
Agentic coding	Claude Code	*Highly specialized*	Codex	*Advanced*
	GitHub Copilot	*Advanced*		
Markup	LaTeX	*Highly specialized*	Markdown	*Advanced*
	HTML/CSS	*Basic*		
Others	Java	*Advanced*	C/C++	*Intermediate*
	Javascript	*Advanced*	Bash	*Intermediate*
	Android	*Basic*	SQLite	*Basic*

Application software

Modeling & Simulation	Mathematica	*Highly specialized*	COMSOL	*Advanced*
	ABAQUS	*Advanced*	SAP2000	*Intermediate*
CAD	AutoCAD	*Highly specialized*	SketchUp	*Advanced*
IDE	VSCode	*Highly specialized*	IntelliJ IDEA	*Advanced*
	Android Studio	*Intermediate*		
Graphics	Inkscape	*Highly specialized*	GIMP	*Intermediate*
Others	Git	*Highly specialized*	GitHub	*Highly specialized*
	Zotero	*Highly specialized*		

Operating systems

Linux	*Highly specialized*	Windows	*Advanced*
Android	*Advanced*	MacOS	*Intermediate*

Languages

- **Italian:** Native language
- **English:** Professional proficiency
- **Chinese:** Duolingo (A1)

Last update: October 15, 2026